

## MORPHOLOGICAL ANALYSIS PIPELINE

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAMESH/OneDrive/Documents/booking/2.py
```

| Word       | Root    | Suffix | Type         | Normalized |
|------------|---------|--------|--------------|------------|
| connected  | connect | ed     | Inflectional | connect    |
| connecting | connect | ing    | Inflectional | connect    |
| connection | connect | ion    | Derivational | connect    |

## MORPHOLOGICAL PARSING MODULE

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAMESH/OneDrive/Documents/booking/2.py
```

| Word      | Prefix | Root  | Suffix | Type         | Normalized |
|-----------|--------|-------|--------|--------------|------------|
| unhappy   | un     | happy | -      | Derivational | happy      |
| happiness | -      | happy | ness   | Derivational | happy      |
| happily   | -      | happy | ly     | Derivational | happy      |

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
```

## STEMMING BASED PREPROCESSING MODULE

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAMESH/OneDrive/Documents/booking/2.py
```

| Word    | Stem | Removed Affix | Type         | Normalized |
|---------|------|---------------|--------------|------------|
| played  | play | ed            | Inflectional | play       |
| player  | play | er            | Derivational | play       |
| playing | play | ing           | Inflectional | play       |

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
```

## FINITE STATE MORPHOLOGICAL PARSER

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAMESH/OneDrive/Documents/booking/2.py
```

| Word    | Transition                           | Pattern   | Root  | Normalized |
|---------|--------------------------------------|-----------|-------|------------|
| writes  | Start -> write -> +s -> Final        | Regular   | write | write      |
| writing | Start -> write -> +ing -> Final      | Regular   | write | write      |
| written | Start -> write -> irregular -> Final | Irregular | write | write      |

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
```

## PORTER STEMMER BASED PREPROCESSING

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAMESH/OneDrive/Documents/booking/2.py
```

| Word       | Rule Applied     | Intermediate | Final Stem |
|------------|------------------|--------------|------------|
| relational | remove 'ational' | relate       | relat      |
| relation   | remove 'ion'     | relate       | relat      |
| relate     | remove 'e'       | relat        | relat      |

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
```